THE ORDINARY PUSHFORWARD COMPUTES THE DERIVED PUSHFORWARD FOR COMPLEXES OF QUASI-COHERENT MODULES ALONG AFFINE MORPHISMS

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Allowed standard prerequisites (editorial).

- The derived pushforward of a complex consisting of f-acyclic modules is computed by the ordinary pushforward termwise. invoked at proof lines 860-861. Reduces to f-acyclicity of quasi-coherent modules.
- Higher direct images of quasi-coherent modules along affine morphisms vanish. invoked at proof lines 861-861. Gives that quasi-coherent modules are f-acyclic, concluding the first proof.
- On affine morphisms the pushforward of quasi-coherent sheaves can be computed on affine covers. invoked at proof lines 862-863. Alternative proof.

Source reference. Original lemma 0G9N: sources/stacks-perfect/pinned-original-source.tex, source lines 731–731. Cited at proof lines 860-860 as the first ingredient of the combination; the f-acyclicity input and conclusion are inside the counted span.

Source reference. Original lemma 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cited at proof lines 861-861 to supply f-acyclicity of quasi-coherent modules; the conclusion is inside the counted span.

Source reference. Original lemma 0DJK: sources/stacks-perfect/pinned-original-source.tex, source lines 562–562. Cited at proof lines 863-863 for the alternative proof; the main combination is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be an affine morphism of schemes. Let $\mathcal{F}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$-modules. Then $f_*\mathcal{F}^\bullet = Rf_*\mathcal{F}^\bullet$.*

Proof. Combine Lemma 0G9N with Cohomology of Schemes, Lemma 01XC. An alternative proof is to work affine locally on S and use Lemma 0DJK. □

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